ROGII - Wellbore Geology Prediction
Team: WGP Diggers

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1. Project Summary

Roughly 10,000 horizontal wells are drilled worldwide every year, yet much of the drilling process still relies on manual interpretation by experts. These operations require immense technical precision, where even small deviations from the target zone can lead to significant resource waste. If the well drifts into less favorable geology, it results in inefficient energy recovery and may require additional corrective measures that increase the overall environmental footprint of the site.

Interpreting the subsurface is challenging because direct measurements are inherently limited. Data from wells, seismic surveys, and logging tools only show part of the picture. Rock layers start out stacked like a layer cake, but can bend or break along faults, making it hard to know exactly where the drill bit sits within the formation. Geologists and engineers analyze incoming data to steer the well, but current analytical tools often struggle to match the nuance of expert interpretation.

Our goal is to develop a machine learning model that predicts the geology encountered along a horizontal wellbore. Our model should identify favorable layers from drilling data and help guide well placement more accurately during operations.

Our solution could help reduce resource waste by minimizing redundant drilling, improve operational safety by better predicting geological hazards, and move the industry toward automated systems that make faster, more consistent, and data-driven decisions.

2. Approach

We will implement and train a bidirectional, encoder-only Transformer in PyTorch that predicts the per-foot change in True Vertical Thickness (TVT) along the lateral and integrates those increments from the last known value, so the loss is placed on absolute TVT and matches the Root Mean Squared Error (RMSE) the competition scores. We will compare this model against three baselines: a persistence predictor that assumes TVT stays flat, a classical sliding cross-correlation alignment between the well and its typewell, and gradient-boosted trees on windowed features. A key part of our approach is a randomized context-split augmentation that hides the known TVT after a random point and supervises only the rest, which turns 773 wells into many training examples and teaches the model to predict from any amount of context, and we will run an ablation that adds the typewell as a second cross-attention stream to test whether learned log correlation helps. We will validate using a well-grouped split and 5-fold GroupKFold scored at the real prediction-start point, since the public leaderboard covers only three wells and is too noisy to trust, and we will analyze training and validation loss curves, overfitting, and error broken down by region and by distance past the prediction point. We expect a naive per-foot regressor without integration or the typewell to under-perform, and we will use these diagnostics to confirm and correct that.

We plan to implement the preprocessing, dataset and windowing pipeline, training loop, evaluation, baselines, and analysis ourselves. We may use standard PyTorch modules such as nn.TransformerEncoder and existing gradient-boosting libraries, but the data representation, the context-split augmentation, the integrate-then-score head, and the experimental pipeline will be written by the team. Simply running a pre-existing model or repository on the dataset will not be sufficient for our project.

3. Resources / Related Work & Papers

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press
- Zhang, C., Bengio, S., Hardt, M., Recht, B., & Vinyals, O. (2017). Understanding deep learning requires rethinking generalization. ICLR
- Bormann, P., Aursand, P., Dilib, F., Dischington, P., & Manral, S. (2020). FORCE 2020 Machine Learning

Contest. www.github.com/bolgebrygg/Force-2020-Machine-Learning-competition

- Vaswani, A., et al. (2017). Attention is all you need. NeurIPS

- Kuvaev, I., Aguilar, R., Granmayeh, J., Holbrook, R., Cruz, M., & Oldacre, A. (2026). ROGII — Wellbore Geology Prediction. www.kaggle.com/competitions/rogii-wellbore-geology-prediction

4. Dataset

The data comprises horizontal well trajectories and vertical reference logs (typewells) used for geological prediction. The dataset is organized into train/ and test/ directories, where each well is identified by a unique 8-character hash.

File and Field Information

- **train/** — Contains the training data. Each well has three associated files:
  - {WELLNAME}__horizontal_well.csv — Trajectory, geological surfaces, and log data.
    - WELLNAME — unique identifier for the well.
    - MD — Measured Depth (ft): The total length of the wellbore from the surface.
    - X — Easting (ft): Spatial coordinate in the horizontal plane.
    - Y — Northing (ft): Spatial coordinate in the horizontal plane.
    - Z — True Vertical Depth (ft): The vertical distance below sea level.
    - ANCC, ASTNU, ASTNL, EGFDU, EGFDL, BUDA — Predicted depth of various geological formations (Training only).
    - TVT — True Vertical Thickness (ft): The manually interpreted geological position for each 1 ft of the lateral well. This is the target variable (Training only).
    - GR — Gamma Ray (API): Log measuring natural radioactivity of the rock.
    - TVT_input — Input Target (ft): A copy of TVT provided as a feature. This column contains NaN values for the evaluation zone.
  - {WELLNAME}__typewell.csv — Vertical reference log for geological correlation.
    - TVT — Vertical Depth Index (ft): Primary depth reference for the vertical log. Corresponds to TVT (geological position) of the associated horizontal well.
    - GR — Gamma Ray (API): The vertical Gamma Ray signature used for correlation.
    - Geology — Formation Label: Categorical label indicating the geological unit (e.g., EGFDL, BUDA).
  - {WELLNAME}.png — Visualization of the well path and geological cross-section.

- **test/** — Contains the evaluation data for about 200 wells. Each well has two associated files:
  - {WELLNAME}__horizontal_well.csv — Trajectory and log data. In these files, the TVT target is hidden (replaced with NaN) in the evaluation zone.
  - {WELLNAME}__typewell.csv — Vertical reference log for the test well.

Dataset Links:
Train https://www.kaggle.com/competitions/rogii-wellbore-geology-prediction/data?select=train
Test https://www.kaggle.com/competitions/rogii-wellbore-geology-prediction/data?select=test

5. Evaluation Plan

Each horizontal well has TVT values that need to be predicted. Prediction quality is measured as the RMSE of all residual error values. RMSE is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

where $\hat{y}$ is the predicted value, y is the original value, and n is the number of rows in the test data.

6. Risks and Scope

High Complexity and Unfamiliarity with Geological Data

The nature of the data is already very complex which might require some domain expertise. None of the group members have any experience in this field, which might have been beneficial when it comes to interpreting some of the predictors and visualizations provided.

Data Limitations

Provided data from wells, seismic surveys, and logging tools only show part of the picture. The real geological formations may not always be accurately reflected in the provided data.

Computational Complexity

Due to group's lack of expertise in this area, the model may require a lot of iterations and modifications which can drastically increase the compute time.

Scope

- Implement an encoder-only Transformer-based architecture.

7. Ethical Implications

Environmental impact of drilling is already a sensitive subject. This project may help with better drill planning which can lead to less waste and increased efficiency. That being said, in these projects, many geologists and engineers involved to help steer the well by reading the incoming data, where many analytical tools struggle to interpret as good as a subject-matter expert. Heavy reliance on ML models might cause environmental disasters.